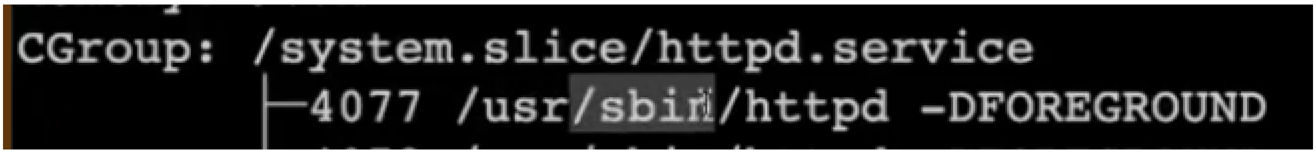
docker_class_5(48)

Using docker file deploy the application

1.
FROM ubuntu
RUN apt update -y
RUN apt install apche2 -y
COPY index.html /var/www/html
CMD ["/usr/sbin/apachectl" , "-D" "FOREGROUND"]

Build and run
docker build -t my-apache .
docker run -d -p 8080:80 --name web1 my-apache


this we call as c group

2.

FROM centos:centos7
LABEL maintainer="Harish"
RUN yum install httpd -y
COPY index.html /var/www/html/
EXPOSE 80
CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]

docker build -t centos-httpd
docker run -d -p 8080:80 --name cweb centos-httpd

3.

FROM rockylinux:9
RUN dnf install -y httpd && dnf clean all
COPY index.html /var/www/html/index.html
EXPOSE 80
CMD ["/usr/sbin/httpd", "-D", "FOREGROUND"]

docker build -t rocky-httpd

docker run -d -p 8080:80 --name rweb rocky-httpd

FROM ubuntu

RUN apt update -y

RUN apt install nginx -y

COPY index.html /var/www/html/

EXPOSE 80

CMD ["nginx", "-g" , "daemon off;"]

docker build -t nginximage .

docker run -itd --name nginxcont -p 8084:80 nginximage

we can do direct base nginx also

5.

FROM nginx

copy index.html /usr/share/nginx/html/

docker build -t nginxbaseimage .

docker run -itd --name nginx-cont -p 8086:80 nginxbaseimage

if we take nginx we use default path is /usr/share/nginx/html/

and we can not start also because already we are using nginx image

```
[root@ip-172-31-15-0 ~]# docker images
REPOSITORY   TAG       IMAGE ID       CREATED          SIZE
flm          1         497298fafef1   6 seconds ago    187MB
myapp        3         d4ec69b21b55   5 minutes ago    179MB
myapp        2         1208bb05cc99   9 minutes ago    466MB
myapp        1         40a433466d50   15 minutes ago   232MB
nginx        latest    c20060033e06   6 days ago       187MB
ubuntu       latest    e4c58958181a   4 weeks ago      77.8MB
centos       centos7   eeb6ee3f44bd   2 years ago      204MB
[root@ip-172-31-15-0 ~]#
```

if you look on this images size is more

in realtime we have to reduce the size in realtime it will take time observer and analysis

if you observe flm has less mb

docker run -itd --name cont6 -p 8086:60 flm:1

as we use nginx we can use httpd also

6.

FROM httpd

COPY index.html /usr/share/apache2/htdocs/

docker build -t flm:2 .

```
[root@ip-172-31-15-0 ~]# docker images
REPOSITORY   TAG       IMAGE ID       CREATED          SIZE
flm          2         1f0000ebac2e   13 seconds ago   168MB
flm          1         497298fafef1   3 minutes ago    187MB
myapp        3         d4ec69b21b55   9 minutes ago    179MB
myapp        2         1208bb05cc99   13 minutes ago   466MB
myapp        1         40a433466d50   19 minutes ago   232MB
nginx        latest    c20060033e06   6 days ago       187MB
httpd        latest    7f6a969e81a5   6 days ago       168MB
ubuntu       latest    e4c58958181a   4 weeks ago      77.8MB
centos       centos7   eeb6ee3f44bd   2 years ago      204MB
[root@ip-172-31-15-0 ~]# docker run -itd --name cont5 -p 8085:80 flm:2
0681f2d5d80c0cc353dee8c7bbb689e414e3bb241c1f724a8d73be0f9ada17b9
```

if you observe flm 2 have very less mb

in realtime how you make less size image

7.

in github clone the staticsite-docker

git clone staticsite-docker

cd /home/harish/docker/staticsite-docker/html

as we have Dockerfile

FROM nginx

COPY . /usr/share/nginx/html

build : docker build -t cycle .

run : docker run -itd --name cont87 -p 7070:80 cycle

it takes very less time

jenkins, ansible it will take more time here no dependency

8.

one more we have digital marketing

clone from github

git clone digital_marketing

cd digital_marketing

cd 2128_tween_agency

here docker file will be there just build

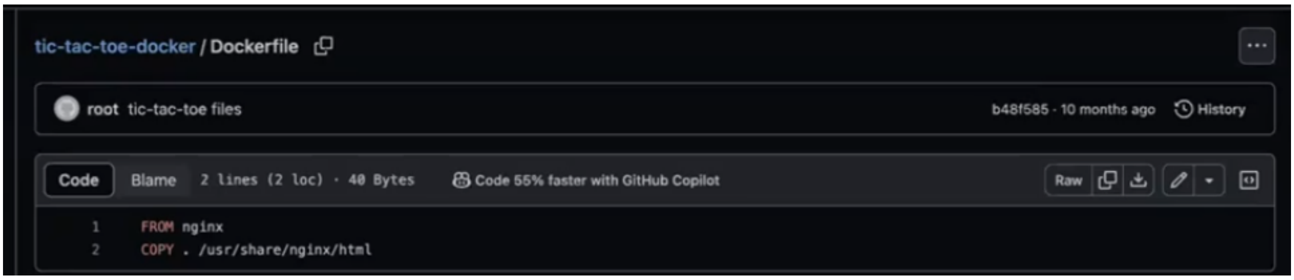
docker build -t website .

docker run -itd --name cont88 -p 9098:80 website

9.

Home work here take this one but take centos image and install nginx and deploy

take this code resource



RUN VS PULL

docker image inspect image_name : it will give information of image

we can do inspect image also

IMAGE LAYERS :

FROM UBUNTU

MAINTAINER NAME HARISH

RUN APT UPDATE -Y

RUN TOUCH HARISH.TXT

RUN APT INSTALL HTTPD -Y

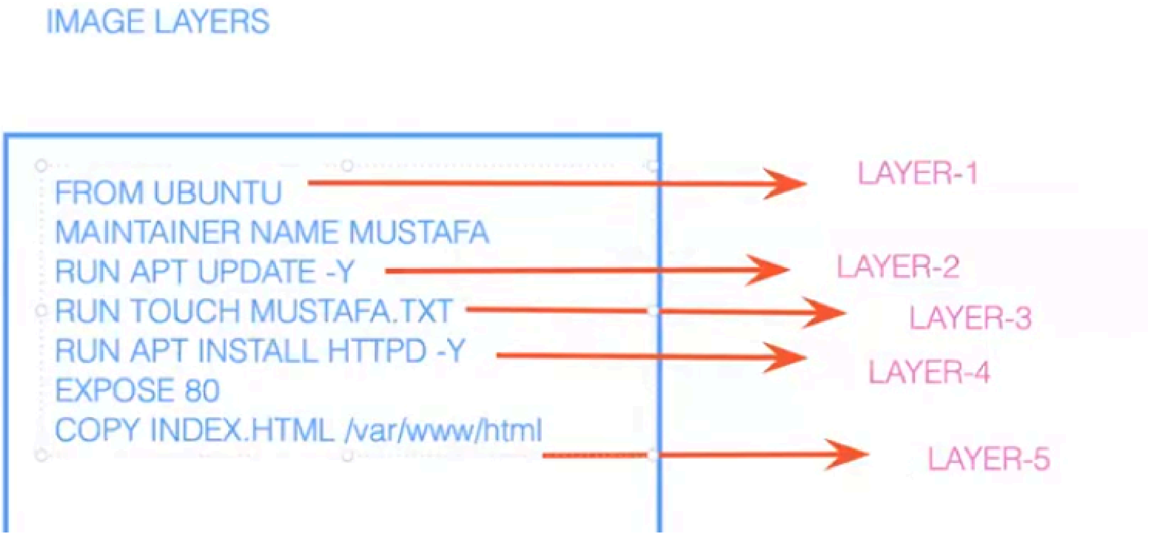
EXPOSE 80

COPY INDEX.HTML /VAR/WWW/HTML

CAN YOU GUESS HOW MANY LAYERS IT WILL FORM : 5 LAYERS

MAINLY IT WILL FOCUS ON WHICH COMMANDS WILL ACTION PERFORM LIKE THAT LAYERS WILL FORM

MAINTAINER AND EXPOSE ONLY DOCUMENT PURPOSE



10.

FROM ubuntu

MAINTAINER harish

RUN apt update -y

RUN touch mustafa.txt

RUN apt install apache2 -y

EXPOSE 80

COPY index.html /var/www/html

#COPY images/ /usr/share/nginx/html/images/

build: docker build -t imagelayer .

check how many layers: docker image inspect imagelayer

```
            "Type": "layers",
            "Layers": [
                "sha256:f2a7f072635332d307212e318e07284948b89f4167fce5c4d7c9cfb7590b74b6",
                "sha256:5f91bf036b7757cd8cd266c2d386833239e26bb624c0440b192460d63b18eae9",
                "sha256:133b80fcb991cf28396b936adeeb5d57a2d3d1abe923e6124db8e775244e0a02",
                "sha256:9cfdf469c934f163b1c237b378d6ba09b0705a696c1ef65ffbacfe498606eab4",
                "sha256:4290339d4dc4e33486e4c0b79ffa133aee614a10f38d2f6b6994d2ae5aa7d3c0"
            ]
        },
```

how many layers are there means : 5

to delete image : delete rmi imagename

to delete all images: docekr rmi $(docker images)

note: before delete image container should be stop

Note: existing image alredy present but again you build the image then the image will overide by privous old goes like none:none type as overide you will see latest

so none images how it will delete means : docker image prune